

U.G.C. NET EX

COMPUTER

(Solved with Expla

1. For the 8-bit word 00111001, the check bits stored with it would be 0111. Suppose when the word is read from memory, the check bits are calculated to be 1101. What is the data word that was read from memory?
- (a) 10011001 (b) 00011001
(c) 00111000 (d) 11000110

Ans : (b) Given 8 bit word 00111001. encoded in d8 = 0, d₇ = 0, d₆ = 1 d₅ = 1, d₄ = 1 d₃ = 0, d₂ = 0, d₁ = 1
parity is calculated as 0111 which are written as p8p4p2p1
p1 = 1 p4 = 1 p2 = 1 p8 = 0
encoded the string as
d8d7d6d5p8d4d3d2p4d1p2p1
001101001111
check bit at other end is 1101
C1 = 1 C4 = 1
C2 = 0 C8 = 1
XOR(1101, 0111) = 1010. Its decimal conversion is 10,
so 10th is changed. 10th digit is d6. so we flip & 4x get
00011001

2. Consider a 32-bit microprocessor, with a 16 - bit external data bus, driven by an 8 MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate for this microprocessor?
- (a) 8×10^6 bytes/sec (b) 4×10^6 bytes/sec
(c) 16×10^6 bytes/sec (d) 4×10^9 bytes/sec

Ans : (b) Since minimum bus cycle duration = 4 clock
cycle bus clock = 8 MHZ
then, maximum bus cycle rate = $8 \text{ m}/4 = 2\text{m/sec}$
Data transferred per bus cycle = 16 bit = 2 B
Data transfer rate per second = bus cycle rate $\times$ data per
bus cycle = $2 \text{ M} \times 2 = 4 \times 10^6 \text{ B/sec}$
So option (b) is correct.

3. The RST 7 instruction in 8085 microprocessor is equivalent to :
- (a) CALL 0010 H (b) CALL 0034 H
(c) CALL 0035 H (d) CALL 003C H

Ans : (c)

Interrupt	Vector Address
RST0	0000 H
RST1	0005 H
RST2	0010 H

Exam. June, 2015
COMPUTER SCIENCE
(Explanation Paper–III)

RST3	0015 H
RST4	0020 H
RST5	0025 H
RST6	0030 H
RST7	0035 H

4. The equivalent hexadecimal notation for octal number 2550276 is :

- (a) FADED (b) AEOBE
(c) ADOBE (d) ACABE

Ans : (c) Given octal No. is
 $(2550276)_8 = (010\ 101\ 101\ 000\ 010\ 111\ 110)_2$
 $\Rightarrow (01\ 010\ 1101\ 0000\ 1011\ 1110)_2$
 $\Rightarrow (0AD0BE)_{16}$
 $(AD0BE)_{16}$

5. The CPU a system having 1 MIPS execution needs 4 machine cycles on an average for executing an instruction. The fifty percent of the cycles use memory bus. A memory read/write employs one machine cycle. For execution of the programs, the system utilizes 90 percent of the CPU time. For block data transfer, an IO device is attached to the system while CPU executes the background programs continuously. What is the maximum I/O data transfer rate if programmed I/O data transfer technique is used?

- (a) 500 Kbytes/sec (b) 2.2 Mbytes/sec
(c) 125Kbytes/sec (d) 250 Kbytes/sec

Ans : (d) CPU speed 1 MPS = 1096 instruction per sec
1 CPU instruction = 4 machine cycle (AVG)
1 memory read/access = 1 machine cycle
CPU utilization = 90%
Programmed I/O
each byte transfer requires 4 cycle (instruction) in status check status
Branch
Read/write data
max data transfer rate $\frac{\text{CPU speed}}{4} = \frac{10^6}{4} = 250\ \text{kbytes/sec}$

6. The number of flip-flops required to design a modulo – 272 counter is :

- (a) 8 (b) 9
(c) 27 (d) 11

Ans : (b) Remember with 'n' number of flip-flop you can get upto maximum modulo - (2^n) counter.
Since $2^9 = 512 > 272$ it can satisfy the condition.

7. Let E_1 and E_2 be two entities in E-R diagram with simple single valued attributes. R_1 and R_2 are two relationships between E_1 and E_2 where R_1 is one - many and R_2 is many - many. R_1 and R_2 do not have any attribute of their own. How minimum number of tables are required to represent this situation in the Relational model?

- (a) 4 (b) 3
(c) 2 (d) 1

Ans : (b) One table for each entity : 2

R_1 is one to many (No separate table): 0

R_2 is many to many and requires one separate table: 1

Total number of tables: 3

8. The STUDENT information in a university stored in the relation STUDENT (Name, SEX, Marks, DEPT_Name)

Consider the following SQL query **SELECT DEPT_Name for STUDENT where SEX = 'M' group by DEPT_Name having avg (Marks) > SELECT avg (Marks) STUDENT. It Returns the Name of the Department for which :**

- (a) The average marks of male students is more than the average marks of students in the same Department
(b) The average marks of male students is more than the average marks of students in the University
(c) The average marks of male students is more than the average marks of male students in the University
(d) The average marks of students is more than the average marks of male students in the University

Ans : (b) The average marks of male students is more than the average marks of student in the university.

9. Select the 'False' statement from the following statements about Normal forms :

- (a) Lossless preserving decomposition into 3NF is always possible
(b) Lossless preserving decomposition into BCNF is always possible
(c) Any Relation with two attributes is in BCNF
(d) BCNF is stronger than 3NF

Ans : (c) A BCNF decomposition which is both lossless as well as dependency preserving at the same time is not always possible.

Example: Relation R(a, b, c) with FDs($ab \rightarrow c$, $c \rightarrow b$) cannot be preserved when decomposition is done.

10. The Relation

Vendor Order (V_no, V_ord_no, V_name, Qty_sup, unit_price)

is in 2NF because :

- (a) Non_key attribute V_name is dependent on V_no which is part of composite key

- (b) Non_key attribute V_name is dependent on Qty_sup
- (c) Key attribute Qty_sup is dependent on primary_key unit price
- (d) Key attribute V_ord_no is dependent on primary_key unit price

Ans : (a) Option (a) is which it is clearly mentioned that partial dependency exists is against 2NF principle. Question seems incomplete, because there is nothing said about functional dependency.

11. The relation schemas R_1 and R_2 form a Lossless join decomposition of R if and only if :

- (1) $R_1 \cap R_2 \twoheadrightarrow (R_1 - R_2)$
- (2) $R_1 \rightarrow R_2$
- (3) $R_1 \cap R_2 \twoheadrightarrow (R_2 - R_1)$
- (4) $R_2 \rightarrow R_1 \cap R_2$

Codes :

- (a) (1) and (2) happens
- (b) (1) and (4) happens
- (c) (1) and (3) happens
- (d) (b) and (3) happens

Ans : (c) Let F be a set of functional dependencies on R, and R_1 and R_2 form a decomposition of R. The decomposition is a lossless join decomposition of R if at least one of the following functional dependencies are in F^+

1- $R_1 \cap R_2 \rightarrow R_1$

2- $R_1 \cap R_2 \rightarrow R_2$

12. In the indexed scheme of blocks to a file, the maximum possible size of the file depends on :

- (a) The number of blocks used for index and the size of index
- (b) Size of Block and size of Address
- (c) Size of index
- (d) Size of Block

Ans:(a) When indexes are created, the maximum number of blocks given to a file depends upon the size of the index which indicates the number of blocks required and size of each block.

13. Give the number of principal vanishing point (s) along their direction for the standard perspective transformation :

- (a) Only one in the direction K
- (b) Two in the directions I and J
- (c) Three in the directions I, J and K
- (d) only two in the directions J and K

Ans : (a) In this case the view plane normal n is vector. Since $N.I = 0$ & $N.J = 0$, so there are no vanishing points in the direction I & J. On the other hand $N.K = K.K = 1$. Thus there is only one principal vanishing point & it is in k direction.

14. Consider a triangle A (0,0), B(1,1) and C(5,2). The triangle has to be rotated by an angle of 45° about the point P(-1, -1). What shall be the coordinates of the new triangle?

- (a) $A' = (1, \sqrt{2} - 1), B' = (-1, 2\sqrt{2} - 1)$
 and $C' = \left(3\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1\right)$
- (b) $A' = (1, \sqrt{2} - 1), B' = (2\sqrt{2} - 1, -1)$
 and $C' = \left(3\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1\right)$
- (c) $A' = (-1, \sqrt{2} - 1), B' = (-1, 2\sqrt{2} - 1)$
 and $C' = \left(3\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1\right)$
- (d) $A' = (\sqrt{2} - 1, -1), B' = (-1, 2\sqrt{2} - 1)$
 and $C' = \left(3\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1\right)$

Ans : (c)

$$R_{45} \quad p = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -1 & (\sqrt{2} - 1) & 1 \end{pmatrix}$$

Now,

$$\begin{bmatrix} A' \\ B' \\ C' \end{bmatrix} = \begin{bmatrix} A \\ B \\ C \end{bmatrix} R_{45}.P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 5 & 2 & 1 \end{bmatrix} \begin{pmatrix} -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -1 & (\sqrt{2} - 1) & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & (\sqrt{2} - 1) & 1 \\ -1 & (2\sqrt{2} - 1) & 1 \\ \left(\frac{3}{2}\sqrt{2} - 1\right) & \left(\frac{9}{2}\sqrt{2} - 1\right) & 1 \end{pmatrix}$$

So $A' = (-1, \sqrt{2} - 1), B' = (-1, 2\sqrt{2} - 1) \&$
 $C' = \left(\frac{3}{2}\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1\right)$

15. The process of dividing an analog signal into a string of discrete outputs, each of constant amplitude, is called :

- (a) Strobing (b) Amplification
 (c) Conditioning (d) Quantization

Ans : (d) Quantization is the process of transforming sampled amplitude values of a message signal into a discrete amplitude value.

- 16. Which of the following is not a basic primitive of the Graphics kernel system (GKS) ?**
- (a) POLYLINE (b) POLYDRAW
(c) FILL AREA (d) POLYMARKER

Ans : (b) The main five primitives in graphics kernel system (GKS)

(1) Polyline : which draw a sequence of connected line segments

(2) Polymarker : Which marks a sequence of points with same symbol.

(3) Fill area : which displays a specified area.

(4) Text : which dreams a string of characters

(5) Cell array : which displays an image composed of a variety of colors or grey scales.

- 17. Which of the following statement(s) is/are incorrect?**
- (1) Mapping the co-ordinates of the points and lines that form the picture into the appropriate co-ordinates on the device or workstation is known as viewing transformation
- (2) The right-handed Cartesian co-ordinates system in whose co-ordinates we describe the picture is known as world co-ordinate system
- (3) The co-ordinate that corresponds to the device or workstation where the image is to be displayed is known as physical device co-ordinate system
- (4) Left - handed co-ordinate system in which the display area of the virtual display device corresponds to the unit $(|x|)$ square whose lower left-hand corner is at the origin of the co-ordinate system, is known as normalized device co-ordinate system
- (a) (1) only (b) (1) and (2)
(c) (3) only (d) (4) only

Ans : (d) Normalized device co-ordinate system make up a coordinate system that describes positions on a virtual plotting device. The left corresponds to (0,0) & the upper right corner corresponds to (1,1)

NDCs can be used when you want to position text, lines, markers or polygons anywhere on the plotting device.

- 18. Match the following :**

List - I		List - II	
(A)	Flood Gun	(i)	An electron gun designed to flood the entire screen with electrons
(B)	Collector	(ii)	Partly energized by flooding gun, stores the charge generated by the

			writing gun
(C)	Ground	(iii)	Used to discharge the collector
(D)	Phosphorus grains	(iv)	Used in memory - tube display and similar to those used in standard CRT
(E)	Writing Gun System	(v)	Used in memory - tube display and basically the same as the electron gun used in a conventional CRT

Code :

	A	B	C	D	E
(a)	(i)	(ii)	(iii)	(iv)	(v)
(b)	(ii)	(iii)	(i)	(iv)	(v)
(c)	(iii)	(i)	(ii)	(v)	(iv)
(d)	(iv)	(v)	(i)	(ii)	(iii)

Ans : (a)

- (a) Flood gun is an electron gun designed to flood the entire screen with electrons, so it satisfies (i)
 (b) Collector are the party energized by flooding gun, stores the charge generated by the writing gun, so it satisfies (ii)
 (c) Ground is used to discharge the collector, so it satisfies (iii)
 (d) Phosphorus grains is used in memory, so it satisfies (iv)
 (e) Writing gun system is also used in memory is used in conventional CRT.

19. Minimal deterministic finite automaton for the language $L = \{0^n \mid n \geq 0, n \neq 4\}$ will have :

- (a) 1 final state among 5 states
 (b) 4 final states among 5 states
 (c) 1 final state among 6 states
 (d) 5 final states among 6 states

Ans : (d)



So final states are $\rightarrow 5$
 non-final state $\rightarrow 1$

20. The regular expression corresponding to the language L where $L = \{x \in \{0, 1\}^* \mid x \text{ ends with } 1 \text{ and does not contain substring } 00\}$ is :

- (a) $(1 + 01)^* (10 + 01)$
 (b) $(1 + 01)^* 01$
 (c) $(1 + 01)^* (1 + 01)$
 (d) $(10 + 01)^* 01$

Ans : (c) Because option (c) will generate all the string ending with 1 & also it will not generate substring as 00.

21. The transition function for the language $L = \{w \mid n_a(w) \text{ and } n_b(w) \text{ are both odd}\}$ is given by :

$$\delta(q_0, a) = q_1 \quad ; \quad \delta(q_0, b) = q_2$$

$$\delta(q_1, a) = q_0 \quad ; \quad \delta(q_1, b) = q_3$$

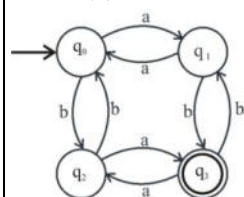
$$\delta(q_2, a) = q_3 \quad ; \quad \delta(q_2, b) = q_0$$

$$\delta(q_3, a) = q_2 \quad ; \quad \delta(q_3, b) = q_1$$

The initial and final states of the automata are:

- (a) q_0 and q_0 respectively
- (b) q_0 and q_1 respectively
- (c) q_0 and q_2 respectively
- (d) q_0 and q_3 respectively

Ans : (d)



State	Description
q_0	a's even, b's even
q_1	a's odd, b's even
q_2	a's even, b's odd
q_3	a's odd, b's odd

Where q_0 as initial state and q_3 accepting state (final state).

22. The clausal form of the disjunctive normal form $\neg A \vee \neg B \vee \neg C \vee D$ is :

- (a) $A \wedge B \wedge C \Rightarrow D$
- (b) $A \vee B \vee C \vee D \Rightarrow \text{true}$
- (c) $A \wedge B \wedge C \wedge D \Rightarrow \text{true}$
- (d) $A \wedge B \wedge C \wedge D \Rightarrow \text{false}$

Ans : (a) We know $A \rightarrow B$ is equivalent to $\sim A \vee B$ now checking option A

$A \wedge B \wedge C \Rightarrow D$ applying about equivalence

$\neg (A \wedge B \wedge C) \vee D$ applying de-morgans law

$\Rightarrow (A \wedge B \wedge C) \rightarrow D$

Hence answer is option (a)

23. Which of the following is false for the programming language PROLOG?

- (a) A PROLOG variable can only be assigned to a value once
- (b) PROLOG is a strongly typed language
- (c) The scope of a variable in PROLOG is a single clause or rule
- (d) The scope of a variable in PROLOG is a single query

Ans : (b) Since prolog is a dynamically typed language. Which is to say that programs are not checked for type errors until run time.

24. Which one of the following is true :

- (a) The resolvent of two Horn clauses is not a Horn clause
- (b) The resolvent of two Horn clauses is a Horn clause
- (c) If we resolve a negated goal G against a fact or rule A to get clause C then C has positive literal or non-null goal
- (d) If we resolve a negated goal G against a fact or rule A to get clause C then C has positive literal or null goal

Ans : (b) If you resolve horn clauses A & B to get clause C, then the positive literal of A will resolve against a negative in B. So the only positive literal left in C is the one, from B (if any), Thus, the resolvent of two horn. Clauses is a horn clause.

25. Which transmission technique guarantees that data packets will be received by the receiver in the same order in which they were sent by the sender?

- (a) Broadcasting
- (b) Unicasting
- (c) Packet switching
- (d) Circuit switching

Ans : (d) Circuit switching transmission technique guarantees the data packets will be received by the receiver in the same order in which they were sent by the sender because path is fixed and all packet goes through that path in same order.

26. Which of the following control fields in TCP header is used to specify whether the sender has no more data to transmit?

- (a) FIN
- (b) RST
- (c) SYN
- (d) PSH

Ans : (a) FIN control fields in TCP header is used to specify whether the sender has no more data to transmit i.e. close the connection from sender side.

27. Which are the two modes of IP security?

- (a) Transport and certificate
- (b) Transport and tunnel
- (c) Certificate and tunnel
- (d) Preshared and transport

Ans : (b) **Transport mode:-** IP header, IPSec, headers IP payload.

Tunnel Mode:- New IP header, IPSec, headers, old IP header, IP payload.

IP Sec can be implemented in a host-to-host transport mode, as well as in network tunneling mode.

28. A message "COMPUTER NETWORK" encrypted (ignore quotes) using columnar transposition cipher with a key "LAYER". The encrypted message is :

- (a) CTTOEWMROPNRUEK
- (b) MROUEKCTTPNROEW
- (c) OEWPNRCTTUEKMRO

(d) UEKPNRMROOEWCCTT

Ans : (c)

3	1	5	2	4
C	O	M	P	U
T	E	R	N	E
T	W	O	R	K

Now, read column by column; first 1st, 2nd, 3rd, 4th then 5th. Therefore, encrypted message is:-
OEW PNR CTT UEK MRO

29. Suppose a digitized voice channel is made by digitizing 8 kHz bandwidth analog voice signal. It is required to sample the signal at twice the highest frequency (two samples per hertz). What is the bit rate required, if it is assumed that each sample requires 8 bits?

- (a) 32 kbps (b) 64 kb
(c) 128 kbps (d) 256 kbps

Ans : (c) Bandwidth of analog channel is 8 k/sec

Required sample 2 samples/nz = 2s

One sample is 8 bit

So bit rate = $(8 \text{ k}) \times 2 \times (8 \text{ bit}) = 128 \text{ kbps}$

30. The maximum payload of a TCP segment is :

- (a) 65, 535 (b) 65, 515
(c) 65, 495 (d) 65, 475

Ans : (d) The maximum payload of TCP segment is 65, 495 bytes. Because maximum IP packet is 65, 535 bytes out of which 20 is used for header. TCP header is 20 bytes.

So, $2^{16} - 20 - 20 = 65535 - 40 = 65495$ bytes left for TCP.

31. An all-pairs shortest-paths problem is efficiently solved using :

- (a) Dijkstra' algorithm
(b) Bellman-Ford algorithm
(c) Kruskal algorithm
(d) Floyd-Warshall algorithm

Ans : (d) Floyd-Warshall_algorithm is used for all-pairs shortest paths problem efficiently with time complexity $O(n^3)$ using programming.

32. The travelling salesman problem can be solved in :

- (a) Polynomial time using dynamic programming algorithm
(b) Polynomial time using branch-and-bound algorithm
(c) Exponential time using dynamic programming algorithm or branch-and-bound algorithm
(d) Polynomial time using backtracking algorithm

Ans : (c) It will be solved by using branch and bound technique. The travelling salesman problem is NP-complete problem. It can also be solved by- one of the earliest application of dynamic programming is the Held-Karp algorithm to solve TSP in time $O(n^2 \cdot 2^n)$.

33. Which of the following is asymptotically smaller?

- (a) $\lg(\lg^* n)$ (b) $\lg^*(\lg n)$
(c) $\lg(n!)$ (d) $\lg^*(n!)$

Ans : (c) Option are in following order:

$$\lg(\lg^* n) < \lg^*(\lg n) < \lg(n!) < \lg^*(n!)$$

34. Consider a hash table of size $m = 100$ and the hash function $h(k) = \text{floor}(m(kA \bmod 1))$ for $A = \frac{(\sqrt{5} - 1)}{2} = 0.618033$. Compute the location to which the key $k = 123456$ is placed in hash table.

- (a) 77 (b) 82
(c) 88 (d) 89

Ans : (c) $n(k) \text{ floor} = (m(kA \bmod 1))$

$$A = \frac{\sqrt{5} - 1}{2} = 0.618023 \quad m = 100$$

if $k = 123456$

then

$$n(k) = \text{floor}(100(123456 \times 0.618023 \bmod 1))$$

$$\Rightarrow \text{floor}(100(76189.88245 \bmod 1))$$

$$= \text{floor}(100(0.882045))$$

$$= \text{floor}(88.2045)$$

$$= 88$$

35. Let $f(n)$ and $g(n)$ be asymptotically non-negative functions. Which of the following is correct?

- (a) $\Theta(f(n) * g(n)) = \min(f(n), g(n))$
(b) $\Theta(f(n) * g(n)) = \max(f(n), g(n))$
(c) $\Theta(f(n) + g(n)) = \min(f(n), g(n))$
(d) $\Theta(f(n) + g(n)) = \max(f(n), g(n))$

Ans : (d) $\Theta(f(n) + g(n)) = \max(f(n), g(n))$

From “asymptotically non-negative”, we can assume that

$$\exists n_1, n_2: f(n) \geq 0 \text{ for } n > n_1$$

$$g(n) \geq 0 \text{ for } n > n_2$$

$$\text{Let } n_0 = \max(n_1, n_2)$$

Some obvious thing for $n > n_0$:

$$f(n) \leq \max(f(n), g(n))$$

$$g(n) \leq \max(f(n), g(n))$$

$$\frac{(f(n) + g(n))}{2} \leq \max(f(n), g(n))$$

$$\max(f(n), g(n)) \leq f(n) + g(n)$$

From the last two inequalities, we get:

$$0 \leq \frac{1}{2}(f(n) + g(n))$$

$$\leq \min(f(n), g(n))$$

$$\leq f(n) + g(n)$$

For $n > n_0$

Which is the definition of $\Theta(f(n) + g(n))$ with $C_1 = 1/2$ and $C_2 = 1$

36. The number of nodes of height h in any n -element heap is

- (a) h (b) z^h
 (c) $\text{ceil}\left(\frac{n}{z^h}\right)$ (d) $\text{ceil}\left(\frac{n}{z^h + 1}\right)$

Ans : (d) **Theorem:** Show that there at most

$\left\lceil \frac{n}{2^{h+1}} \right\rceil$ nodes of height h in any n -element heap.

Proof:

(i) **Base:** $h = 0$. The number of leaves is

$$\left\lceil \frac{n}{2} \right\rceil = \left\lceil \frac{n}{2^{0+1}} \right\rceil$$

(ii) **Step:** Let's assume it holds for nodes of height $(h - 1)$. Let's a tree and remove all its leaves.

We get a new tree with $\left(n - \left\lceil \frac{n}{2} \right\rceil\right) = \left\lceil \frac{n}{2} \right\rceil$ elements. Note

that the nodes with height h in the old tree have height $(h - 1)$ in the new one. We will calculate the number of such nodes in the new tree. By the inductive assumption we have that T , the number of nodes with height $(h - 1)$ in the new tree is:

$$\left\lceil \left\lceil \frac{n}{2} \right\rceil / 2^{h-1+1} \right\rceil < \left\lceil \frac{n/2}{2^h} \right\rceil = \left\lceil \frac{n}{2^{h+1}} \right\rceil$$

As mentioned, this is also the number of nodes with height h in the old tree.

This question has typing of options error, so UGC gives marks to all.

37. In Java, when we implement an interface method, it must be declared as :

- (a) Private (b) Protected
 (c) Public (d) Friend

Ans : (c) **Interface:-** It is used to provide total abstraction. That means all the methods in interface are declared with empty body and are public and all fields are public, static and final default.

38. The Servlet Response interface enables as servlet to formulate a response for a client using the method

- (a) `void log (Exception e, String s)`
 (b) `void destroy`
 (c) `int get Server port ()`
 (d) `void set Context type (string type)`

Ans : (*) The Servlet Response object enables the servlet to formulate a response for the client using the method get server name () get write (), and Set Content Type ().

39. Which one of the following is correct?

- (a) Java applets can be written in any programming language
- (b) An applet is not a small program
- (c) An applet can be run on its own
- (d) Applets are embedded in another applications

Ans : (d) An applet is a small program that is intended not to be run on its own, but rather to be embedded inside-another application.

40. In XML we can specify the frequency of an element by using the symbols :

- (a) + * !
- (b) # * !
- (c) + * ?
- (d) - * ?

Ans : (c) + symbol for one or more time
* for 0 or more time? zero or one time.

41. In XML, DOCTYPE declaration specifies to include a reference to file.

- (a) Document type Definition
- (b) Document type declaration
- (c) Document transfer definition
- (d) Document type language

Ans : (a) In XML, DOCTYPE declaration can be used to define special characters & character string used in the document & it include a reference of document type definition file

42. Module design is used to maximize cohesion and minimize coupling. Which of the following is the key to implement this rule?

- (a) Inheritance
- (b) Polymorphism
- (c) Encapsulation
- (d) Abstraction

Ans : (c) We know that the golden rule in module design is used to maximize cohesion and minimize coupling.

- Encapsulation is the key to implement this rule

43. Verification :

- (a) refers to the set of activities that ensure that software correctly implements a specific function
- (b) gives answer to the question - Are we building the product right?
- (c) requires execution of software
- (d) both (a) and (b)

Ans : (d) Verification means asking "are we building the product right" verification refers to the set of activities that ensure software correctly implements the specific function.

- 44. Which design metric is used to measure the compactness of the program in terms of lines of codes ?**
- (a) Consistency (b) Conciseness
 - (c) Efficiency (d) Accuracy

Ans : (a) Consistency:- The use of uniform design and documentation techniques through the software development project.

- 45. Requirements prioritization and negotiation belongs to :**
- (a) Requirements validation
 - (b) Requirements elicitation
 - (c) Feasibility study
 - (d) Requirements reviews

Ans : (b) Requirements elicitation.

- 46. Adaptive maintenance is a maintenance which**
- (a) correct errors that were not discovered till testing phase
 - (b) is carried out to port the existing software to a new environment
 - (c) improves the system performance
 - (d) both (b) and (c)

Ans : (b) Maintenance is a modification of a software product performed after delivery to keep a software product usable in a changed or changing environment.

- 47. A Design concept refinement is a :**
- (a) Top-down approach
 - (b) Complementary of abstraction concept
 - (c) Process of elaboration
 - (d) All of the above

Ans : (d) Refinement is surly a top down approach as we are extracting a particular type like person entity to s/w engineer.

Refinement, process of elaboration, start with the statement of function defined at the abstract level, decompose the statement function in a step wise fashion until programming language statements are reached.

- 48. A software design is highly modular if :**
- (a) cohesion is functional and coupling is data type
 - (b) cohesion is coincidental and coupling is data type
 - (c) cohesion is sequential and coupling is content type
 - (d) cohesion is functional and coupling is stamp type

Ans : (a) We know our aim is to maximize cohesion & minimum coupling

Strong form of cohesion is functional cohesion weakest form of coupling is data coupling.

49. Match the following for operating system techniques with the most appropriate advantage :

List - I		List - II	
(A)	Spooling	(i)	Allows several jobs in memory to improve CPU utilization
(B)	Multiprogramming	(ii)	Access to shared resources among geographically dispersed computers in a transparent way
(C)	Time sharing	(iii)	Overlapping I/O and computations
(D)	Distributed computing	(iv)	Allows many users to share a computer simultaneously by switching processor frequently

Codes :

	A	B	C	D
(a)	(iii)	(i)	(ii)	(iv)
(b)	(iii)	(i)	(iv)	(ii)
(c)	(iv)	(iii)	(ii)	(i)
(d)	(ii)	(iii)	(iv)	(i)

Ans : (b) (a) Spooling is overlapping I/O & computation (iii)
 (b) Multiprogramming refers to allows several job in memory for improving the CPU utilization (i)
 (c) Time sharing is allow to many user to share a computer simultaneously by switching processor frequently (iv)
 (d) Distributed computing Access to shared resource among geographically dispersed computers in a transparent way (ii)

50. Which of the following statements is not true for multi level feedback queue processor scheduling algorithm?

- (a) Queues have different priorities
- (b) Each queue may have different scheduling algorithm
- (c) Processes are permanently assigned to a queue
- (d) This algorithm can be configured to match a specific system under design

Ans : (c) Processes are not permanently assigned to a queue they can vary between queues as their priorities change
 So rest (3) option are true and the answer is option (c)

- 51. What is the most appropriate function of Memory management Unit (MMU)?**
- (a) It is an associative memory to store TLB
 - (b) It is a technique of supporting multiprogramming by creating dynamic partitions
 - (c) It is a chip to map virtual address to physical address
 - (d) It is an algorithm to allocate and deallocate main memory to a process

Ans : (c) A memory management unit (MMU), sometimes called paged memory management unit (PMMU), is a computer hardware unit having all memory references passing through itself, primarily performing the translation of virtual memory addresses to physical addresses. It is usually implemented as part of the central processing unit (CPU), but it also can be in the form of a separate integrated circuit. So we can say it is a chip to map virtual address to physical address

- 52. Dining philosopher's problem is a :**
- (a) Producer - consumer problem
 - (b) Classical IPC problem
 - (c) Starvation problem
 - (d) Synchronization primitive

Ans : (b) The dining philosophers problem is an example problem often used in concurrent algorithm design to illustrate synchronization issues and techniques for resolving them. It's a classical IPC problem.

- 53. In allocation method for disk block allocation in a file system, insertion and deletion of blocks in a file is easy.**
- (a) Index
 - (b) Linked
 - (c) Contiguous
 - (d) Bit Map

Ans : (b) Just like linked list where insertion, deletion of blocks is easy. In linked allocation only pointer needs to be changed. In linked allocation each file is a linked list of disk blocks. The directory contains a pointer to the first block of the file, for example, a file of 5 blocks which starts at block 4, might continuous at block 7, then block 16, block 10, 8 finally block 27. Each block contains a pointer to the next block & the last block contains a NIL-pointer. The value-1 may be used for NIL to differential it from block 0.

- 54. A Unix file may be of the type :**
- (a) Regular file
 - (b) Directory file
 - (c) Device file
 - (d) Any one of the above

Ans : (d) A Unix file may be of any of the type like regular file, directory file, or device file.

55. Match the following :

List - I		List – II	
(A)	Intelligence	(i)	Contextual, tacit, transfer needs learning
(B)	Knowledge	(ii)	Scattered facts, easily transferable
(C)	Information	(iii)	Judgmental
(D)	Data	(iv)	Codifiable, endorsed with relevance and purpose

Codes :

	A	B	C	D
(a)	(iii)	(ii)	(iv)	(i)
(b)	(iii)	(i)	(iv)	(ii)
(c)	(i)	(ii)	(iii)	(iv)
(d)	(i)	(iii)	(iv)	(ii)

Ans : (b) (i) Intelligence: judgmental
(ii)- Knowledge: contractual tacit, transfer needs learning.
(iii) Information:- Codifiable, endorsed with relevance and purpose.
(iv) Data- Scattered facts, easily transferrable.

56. Match the following knowledge representation techniques with their applications :

List - I		List – II	
(A)	Frames	(i)	Pictorial representation of objects, their attributes and relationships
(B)	Conceptual dependencies	(ii)	To describe real world stereotype events
(C)	Associative networks	(iii)	Record like structures for grouping closely related knowledge
(D)	Scripts	(iv)	Structures and primitives to represent sentences

Codes :

	A	B	C	D
(a)	(iii)	(iv)	(i)	(ii)
(b)	(iii)	(iv)	(ii)	(i)

- (c) (iv) (iii) (i) (ii)
 (d) (iv) (iii) (ii) (i)

Ans : (a)

- (a) Frames are record like structures for grouping closely related knowledge (iii)
 (b) Conceptual dependencies are structure to represent sentence (iv)
 (c) Associative networks are pictorial representation of objects (i)
 (d) Scripts are description of real word stereo type events (ii)

57. In propositional logic $P \leftrightarrow Q$ is equivalent to (where $\sim$ denotes NOT) :

- (a) $\sim (P \vee Q) \wedge \sim (Q \vee P)$
 (b) $(\sim P \vee Q) \wedge (\sim Q \vee P)$
 (c) $(P \vee Q) \wedge (Q \vee P)$
 (d) $\sim (P \vee Q) \rightarrow \sim (Q \vee P)$

Ans : (b) $P \leftrightarrow Q$

$(P \rightarrow Q) \wedge (Q \rightarrow P)$

$(\sim P \vee Q) \wedge (\sim Q \vee P)$

58. Which of the following statements is true for branch - and - bound search?

- (a) Underestimates of remaining distance may cause deviation from optimal path
 (b) Overestimates can't cause right path to be overlooked.
 (c) Dynamic programming principle can be used to discard redundant partial paths.
 (d) All of the above

Ans : (c) Branch & bound is an algorithm design paradigm for discrete & combinational optimization problems branch & bound algorithm consist of a systematic enumeration of candidate solution by means of state space search. A branch & bound is a dynamic programming principle can be used to discard redundant partial paths.

59. Match the following with respect to heuristic search techniques :

List – I		List – II	
(A)	Steepest - gradient Hill climbing	(i)	Keeps track of all partial paths which can be candidate for further exploration
(B)	Branch - and - bound	(ii)	Discover problem state (s) that satisfy a set of constraints

(C)	Constraint satisfaction	(iii)	Detects difference between current state and goal state
(D)	Means end analysis	(iv)	Considers all moves from current state and selects best move

Codes :

	A	B	C	D
(a)	(i)	(iv)	(iii)	(ii)
(b)	(iv)	(i)	(ii)	(iii)
(c)	(i)	(iv)	(ii)	(iii)
(d)	(iv)	(ii)	(i)	(iii)

Ans : (b) (a) Steepest accent hill climbing looks for best more which may or may not be optimal, so it satisfies (iv)
 (b) Branch & bound is an algorithm design for discrete problems & keep tracks of all partial path which can be candidate for further exploration (i)
 (c) Constraint satisfaction (discover problem state is) that satisfy a set of constraints (ii)
 (d) Means end analysis detects difference between current state & goal state.

60. Match the following for methods of MIS development :

List - I		List – II	
(A)	Joint Application Design (JAD)	(i)	Delivers functionality in rapid iteration measured in weeks and needs frequent communication, development, testing and delivery
(B)	Computer aided software Engg	(ii)	Reusable applications generally with one specific function. It is closely linked with idea of web services and service oriented architecture
(C)	Agile development	(iii)	Tools to automate many tasks of SDLC

(D)	Component based technology	(iv)	A group based tool for collecting user requirements and creating system design. Mostly used in analysis and design stages of SDLC
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Codes :

	A	B	C	D
(a)	(i)	(iii)	(ii)	(iv)
(b)	(iv)	(iii)	(i)	(ii)
(c)	(iii)	(iv)	(i)	(ii)
(d)	(iii)	(i)	(iv)	(ii)

Ans : (b) JAD → is a methodology that involves the client or end user in the design and development of an application.

case tools → tools to automate many tasks of SDLC

Agile development → Agile SDLC model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software product.

component based technology – Reusable applications generally with one specific function. It is closely linked with idea of web services.

61. A context free grammar for $L = \{w \mid n_0(w) > (w)\}$ is given by :

- (a) $S \rightarrow 0 \mid 0S \mid 1SS$
- (b) $S \rightarrow 0S \mid 1S \mid 0SS \mid 1SS \mid 0 \mid 1$
- (c) $S \rightarrow 0 \mid 0S \mid 1SS \mid 1SS \mid S1S \mid SS1$
- (d) $S \rightarrow 0S \mid 1S \mid 0 \mid 1$

Ans : (c) (1) $S \rightarrow 0 \mid 0S \mid 1SS$ (It is false since 10001 cant generate)

(2) $S \rightarrow 0S \mid 1S \mid 0SS \mid 1SS \mid 0 \mid 1$ (It is also false since generates 1)

(3) $S \rightarrow 0 \mid 0S \mid 1SS \mid S1S \mid SS1$ (True)

(4) $S \rightarrow 0S \mid 1S \mid 0 \mid 1$ (False since $1 \wedge +$ generated)

62. Given the following two statements :

S1 : If L_1 and L_2 are recursively enumerable languages over Σ , then $L_1 \cup L_2$ and $L_1 \cap L_2$ are also recursively enumerable.

S2 : The set of recursively enumerable languages is countable.

Which of the following is correct?

- (a) S1 is correct and S2 is not correct
- (b) S1 is correct and S2 is correct
- (c) Both S1 and S2 are not correct
- (d) Both S1 and S2 are correct

Ans : (d) **S₁: True:** because recursively enumerable language are closed under union and intersection operation.

S₂: True: the set of all language on Σ is not countable, but the set of all turing machines can be enumerated. So, the set of all recursively enumerable language is countable.

63. Given the following grammars :

G1 : $S \rightarrow AB \mid aaB$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow bB \mid \epsilon$

G2 : $S \rightarrow A \mid B$

$A \rightarrow aAb \mid ab$

$B \rightarrow a b B \mid \epsilon$

Which of the following is correct ?

- (a) G1 is ambiguous and G2 is unambiguous grammars
- (b) G1 is unambiguous and G2 is ambiguous grammars
- (c) both G1 and G2 are ambiguous grammars
- (d) both G1 and G2 are unambiguous grammars

Ans : (c) Both are ambiguous since 1st for aa & second for ab has two derivation tree.

64. Given the symbols A, B, C, D, E, F, G and H

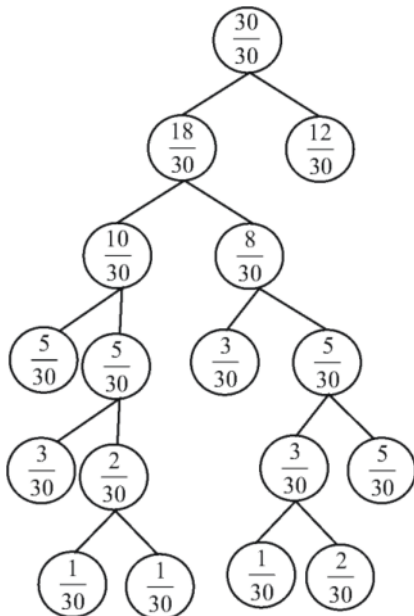
with the probabilities $\frac{1}{30}, \frac{1}{30}, \frac{1}{30}, \frac{2}{30}, \frac{3}{30}, \frac{5}{30},$

$\frac{5}{30}$ and $\frac{12}{30}$ respectively. The average Huffman

code size in bits per symbol is :

- (a) $\frac{67}{30}$
- (b) $\frac{70}{34}$
- (c) $\frac{76}{30}$
- (d) $\frac{78}{30}$

Ans : (c)



$$\Rightarrow 5 \times 1 + 5 \times 1 + 4 \times 1 + 4 \times 2 + 4 \times 3 + 3 \times 5 + 3 \times 5 + 12 \left(\frac{1}{30} \right)$$

$$\Rightarrow 5 + 5 + 4 + 8 + 12 + 15 + 15 + 12 \left(\frac{1}{30} \right)$$

$$= 76$$

So answer is $\left(\frac{76}{30} \right)$

65. The redundancy in images stems from :

- (a) pixel decor relation
- (b) pixel correlation
- (c) pixel quantization
- (d) image size

Ans : (b) The redundancy in image stems from pixel correlation. If we transform the image a representation where the pixels are corrected, we have eliminated the redundancy & the image has been fully compressed.

66. In a binary hamming code the number of check digits is r then number of message digits is equal to :

- (a) $2^r - 1$
- (b) $2^r - r - 1$
- (c) $2^r - r + 1$
- (d) $2^r + r - 1$

Ans : (b) Should be answer $\Rightarrow 2^r - r - 1$

67. In the Hungarian method for solving assignment problem, and optimal assignment requires that the maximum number of lines that can be down through squares with zero opportunity cost be equal to the number of :

- (a) rows of columns
- (b) row + columns
- (c) rows + columns - 1
- (d) rows+columns +1

Ans : (a) Maximum number of lines that can be drawn through squares with zero opportunity cost be equal to the number of rows or columns.

68. Consider the following transportation problem :

		→ Warehouse			
↓ Factory		W ₁	W ₂	W ₃	Supply
	F ₁	16	20	12	200
	F ₂	14	8	18	160
	F ₃	26	24	16	90
	Demand	180	120	150	

The initial basic feasible solution of the above transportation problem using Vogel's Approximation method (VAM) is given below :

→ Warehouse					
↓ Factor y		W ₁	W ₂	W ₃	Supply
	F ₁	16(140)	20	12(60)	200
	F ₂	14(40)	8(120)	18	160
	F ₃	26	24	16(90)	90
	Demand	180	120	150	

The solution of the above problem :

- (a) is degenerate solution
- (b) is optimum solution
- (c) needs to improve
- (d) is infeasible solution

Ans : (b)

w ₁	w ₂	w ₃	
16(140)	20	12(60)	u ₁ = 0
14(40)	8(120)	18	u ₂
26	24	16(90)	u ₃
v ₁	v ₂	v ₃	
c ₁₁ = v ₁ + v ₁		v ₁ = 16	u ₁ = 0
16 = 0 + v ₁		v ₂ = 10	u ₂ = -2
		v ₂ = 12	u ₃ = 4
C ₁₃ = u ₁ + u ₃			
12 = 0 + v ₃			
C ₂₁ = u ₂ + v ₁			
u ₁ = u ₂ + 16			
u ₂ = -2			
C ₂₂ = u ₂ + v ₂			
8 = -2 + v ₂			
v ₂ = 10			
C ₃₃ = u ₃ + v ₃			
16 = u ₃ + 12			
v ₃ = 4			
Un allocated			
d ₁₂ = C ₁₂ - (u ₁ + v ₂)			
= 20 - (0 + 10)			
= 10			
d ₂₃ = (23 - (u ₃ + v ₃))			
= 18 - (-2 + 12)			
= .8			
d ₃₁ = C ₃₁ - (u ₃ + v ₁)			
= 26 - (4 + 16)			
= 6			
d ₃₂ = C ₃₂ - (u ₃ + v ₂)			
= 24 - (4 + 10)			
= 10			

all d's are p + vc so it is optimum solution.

69. Given the following statements with respect to linear programming problem :

S1 : The dual of the dual linear programming problem is again the primal problem

S2 : If either the primal or the dual problem has an unbounded objective function value, the other problem has no feasible solution

S3 : If either the primal or dual problem has a finite optimal solution, the other one also possesses the same, and the optimal value of the objective functions of the two problems are equal.

Which of the following is true ?

- (a) S1 and S2
- (b) S1 and S3
- (c) S2 and S3
- (d) S1, S2 and S3

Ans : (d) All 3 statement are correct as they all properties of dual & primal.

70. Consider the two class classification task that consists of the following points :

Class C1 : [1 1.5] [1 -1.5]

Class C2 : [-2 2.5] [-2 -2.5]

The decision boundary between the two classes using single perceptron is given by :

- (a) $x_1 + x_2 + 1.5 = 0$
- (b) $x_1 + x_2 - 1.5 = 0$
- (c) $x_1 + 1.5 = 0$
- (d) $x_1 - 1.5 = 0$

Ans : (c) Simply plot the all points of C_1 & C_2 in 2-D coordinates then see which option's line separates the points of C_1 & C_2
all points of C_1 should be one side & all points of C_2 should be on other side of line.
only line given in option C $X_1 + 1.5 = 0$ does that.

71. Let A and B be two fuzzy integers defined as :

A = {(1, 0.3), (2, 0.6), (3, 1), (4,0.7), (5,0.2)}

B = {(10, 0.5), (11,1), (12,0.5)}

Using fuzzy arithmetic operation given by

$$\mu_{A+B}^{(Z)} = \bigoplus_{x+y=Z} (\mu_A(x)\mu_B(y))$$

**f(A + B) is [Note: $\oplus = \max$
 $\ominus = \min$]**

- (a) $\{(11, 0.8), (13, 1), (15, 1)\}$
 (b) $\{(11, 0.3), (12, 0.5), (13, 1), (14, 1), (15, 1), (16, 0.5), (17, 0.2)\}$
 (c) $\{(11, 0.3), (12, 0.5), (13, 0.6), (14, 1), (15, 1), (16, 0.5), (17, 0.2)\}$
 (d) $\{(11, 0.3), (12, 0.5), (13, 0.6), (14, 1), (15, 0.7), (16, 0.5), (17, 0.2)\}$

Ans : (d) $A = \{(1, 0.3), (2, 0.6), (3, 1), (4, 0.7), (5, 0.2)\}$

$B = \{(10, 0.5), (11, 1), (12, 0.5)\}$

$A + B (z) = \oplus x + y = z$ ($uA(x) \otimes uB(y)$) is also given as below (add elements of A & B while keeping the min. membership value e.g. $1 + 10 = 11$ & $\min(0.3, 0.5) = 0.3$ & so on for rest

$\oplus x + y = z \{(11, 0.3), (12, 0.3), (13, 0.3), (12, 0.5), (13, 0.6), (14, 0.5), (13, 0.5), (14, 1), (15, 0.5), (14, 0.5), (15, 0.7), (16, 0.5), (15, 0.2), (16, 0.2), (17, 0.2)\}$ bin considering min membership value

Now considering min membership value

$\{(11, 0.3), (15, 0.5), (13, 0.6), (14, 1), (15, 0.7), (16, 0.5), (17, 0.2)\}$

72. Suppose the function y and a fuzzy integer number around - 4 for x are given as $y = (x - 3)^2 + 2$

Around - 4 = $\{(2, 0.3), (3, 0.6), (4, 1), (5, 0.6), (6, 0.3)\}$ respectively. Then f (Around -4) is given by :

- (a) $\{(2, 0.6), (3, 0.3), (6, 1), (11, 0.3)\}$
 (b) $\{(2, 0.6), (3, 1), (6, 1), (11, 0.3)\}$
 (c) $\{(2, 0.6), (3, 1), (6, 0.6), (11, 0.3)\}$
 (d) $\{(2, 0.6), (3, 0.3), (6, 0.6), (11, 0.3)\}$

Ans : (c) $y = (x - 3)^2 + 2$

around- 4 = $\{(2, 0.3), (3, 0.6), (4, 1), (5, 0.6), (6, 0.3)\}$

putting $x = 2, 3, 4, 5, 6$ in $y = (x - 3)^2 + 2$ will give 3 2 3 6 11 with their original x membership values (.3, .6, 1, .6, .3) here is present with 2 membership values .3 & 1 so 1 is considered & final result will be $\{(3, 1), (2, .6), (6, .6), (11, .3)\}$

73. Match the following for Unix system calls :

List - I		List - II	
(A)	exec	(i)	Creates a new process
(B)	brk	(ii)	Invokes another program overlaying memory space with a copy of an executable file

(C)	wait	(iii)	To increase or decrease the size of data region
(D)	fork	(iv)	A process synchronizes with termination of child process

Codes :

	A	B	C	D
(a)	(ii)	(iii)	(iv)	(i)
(b)	(iii)	(ii)	(iv)	(i)
(c)	(iv)	(iii)	(ii)	(i)
(d)	(iv)	(iii)	(i)	(ii)

Ans : (a) exec : replaces the current program in the current process, without forking a new process.

⇒ brk() & sbrk() change the location of the program break, which defines the end of the process's data segment.

⇒ The parent process may then issue a wait system call, which suspends the execution of the parent process while the child executes.

⇒ Fork—creates new process.

74. WOW32 is a :

- (a) Win 32 API library for creating processes and threads
- (b) Special kind of file system to the NT name space
- (c) Kernel - mode objects accessible through Win 32 API
- (d) Special execution environment used to run 16 bit Window applications on 32-bit machines

Ans : (d) Wow 32 is a running environment used in windows to run 16 bit windows application on a 32 bit machine

75. The Unix command :

\$ vi file1 file2

- (a) Edits file1 and stores the contents of file1 in file2
- (b) Both files i.e. file1 and file2 can be edited using 'ex' command to travel between the files
- (c) Both files can be edited using 'mv' command to move between the files
- (d) Edits file1 first, saves it and then edits file2

Ans : (b) You can edit more than one file at a time with vi.

vi file1 file2 vi two (or more) files at the same time.